PassFrame tuition

Simple Algebra

Year 4 Pt 2 — 11+ Lesson Worksheet (PassFrame Tuition)

Worked examples — tutor talks through these first

A) Easy: Find the missing number: 7 + __ = 12.
Method & answer: 12 − 7 = 5. Answer: 5.

B) Standard: Solve: 3x + 4 = 19.
Method & answer: 3x = 15, so x = 5.

C) Advanced: Solve: 2(x + 3) = 20.
Method & answer: x + 3 = 10, so x = 7.

Easy / Standard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. Find the missing number: 5 + __ = 11.

2. Find the missing number: __ − 4 = 9.

3. Find the missing number: 3 × __ = 18. A) 5 B) 6 C) 7 D) 9

4. Find the missing number: __ ÷ 5 = 4.

5. Find the missing number: 20 − __ = 8.

6. Select ALL of these where the missing number is 6: A) __ + 9 = 15 B) __ − 2 = 8 C) 2 × __ = 12 D) 15 ÷ __ = 3 E) __ + 1 = 7

7. Find the missing number: 6 × __ = 30. A) 4 B) 5 C) 6 D) 7

8. Find the missing number: __ ÷ 3 = 7.

9. Find the missing number: 14 + __ = 20.

10. Select ALL of these where the missing number is 12: A) __ − 7 = 5 B) __ + 3 = 16 C) 3 × __ = 36 D) __ ÷ 4 = 3 E) 20 − __ = 8

11. Find the missing number: 4 × __ = 28.

12. Find the missing number: __ ÷ 6 = 3.

13. Find the missing number: 25 − __ = 17.

14. A function machine adds 5, then doubles. If the output is 16, what was the input? A) 3 B) 5 C) 6 D) 8

15. Find the missing number: __ + 15 = 22.

Standard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. Solve: x + 5 = 12.

2. Solve: x − 6 = 10.

3. Solve: 2x = 18. A) 7 B) 8 C) 9 D) 10

6. Select ALL of the equations below where x = 7: A) 2x + 5 = 19 B) 3x − 1 = 20 C) x + 9 = 16 D) 4x = 32 E) 5x − 3 = 32

7. Solve: 5x − 4 = 26.

8. If x = 4, find the value of 3x + 2. A) 12 B) 13 C) 14 D) 15

11. Select ALL of the equations below where x = 4: A) 3x − 7 = 5 B) 2x + 1 = 9 C) 5x = 25 D) x + 10 = 14 E) 6x − 3 = 21

12. Think of a number, double it, then add 6. The answer is 20. What was the number?

13. Solve: 6x + 2 = 26. A) 3 B) 4 C) 5 D) 6

4. Solve: 3x + 1 = 16.

5. Solve: 4x − 3 = 21.

9. If y = 6, find the value of 2y − 5.

10. Solve: x/2 + 3 = 9.

14. If a = 5 and b = 3, find the value of a + 2b.

15. Solve: 10 − x = 4.

Standard / Hard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. Solve: 2(x + 5) = 22.

2. Solve: 3(x − 2) = 21.

3. Solve: 4x + 3 = 2x + 11. A) 3 B) 4 C) 5 D) 6

4. Solve: 5x − 2 = 3x + 12.

5. Solve: 2(x + 1) + 3 = 19.

6. I think of a number, add 7, then multiply by 3. The result is 36. What is the number?

7. Select ALL of the equations below where x = 5: A) 7x − 5 = 4x + 10 B) 3x + 2 = 17 C) 4x − 3 = 2x + 8 D) 2x + 6 = 20 E) 5x − 10 = 15

8. The perimeter of a rectangle is 2(l + w). If the perimeter is 30cm and the width is 6cm, find the length. A) 7cm B) 8cm C) 9cm D) 10cm

9. Solve: 3(2x − 1) = 33.

10. Two numbers have a sum of 24, with one number x and the other x + 6. Select ALL of the statements below that are true: A) x = 9 B) the two numbers are 9 and 15 C) x = 12 D) the larger number is 15 E) the smaller number is 10

11. Solve: 6x + 4 = 4x + 18.

12. A number tripled and then decreased by 5 gives 22. Find the number.

13. Solve: 2x − 3 = −x + 18. A) 5 B) 6 C) 7 D) 8

14. The sum of three consecutive numbers is 39. Find the smallest number.

15. Solve: 4(x − 3) = x + 6.